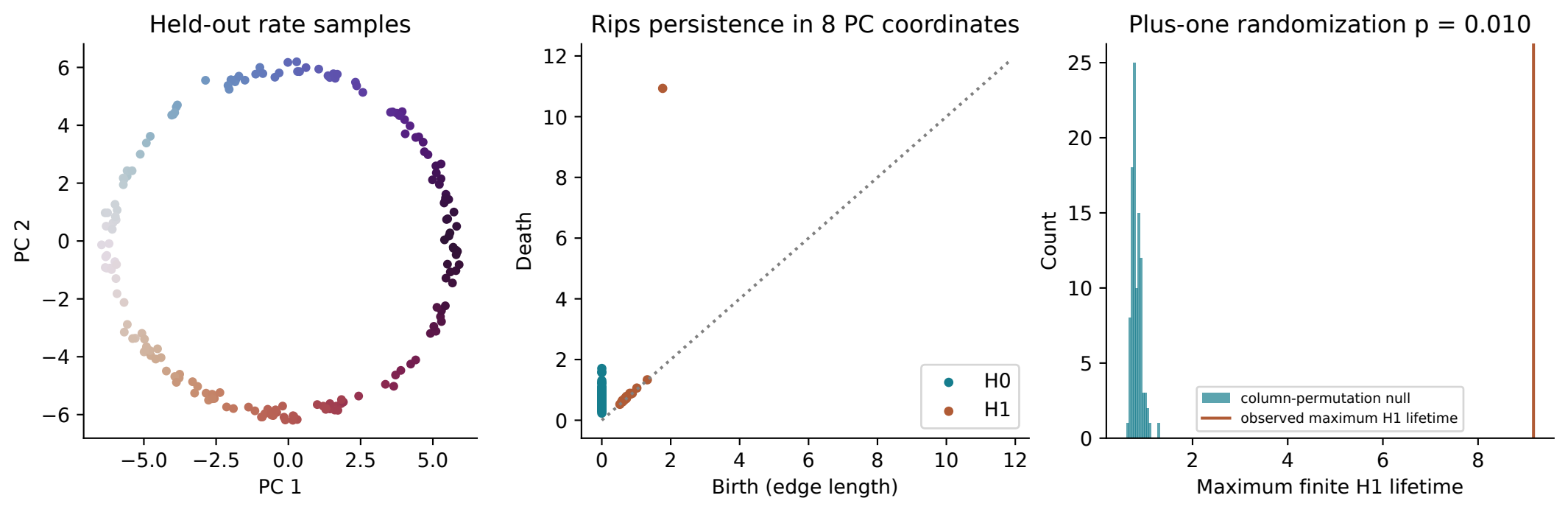
Held-out rate samples
PC 2
PC 1
6
4
2
0
−2
−4
−6
−5.0
−2.5
0.0
2.5
5.0
Rips persistence in 8 PC coordinates
Death
Birth (edge length)
12
10
8
6
4
2
0
H0
H1
Plus-one randomization p = 0.010
Count
Maximum finite H1 lifetime
25
20
15
10
5
0
column-permutation null
observed maximum H1 lifetime